

Smart Health Web Dashboard Requirement Specification

Frontend Architecture, Component Hierarchy & Feature Specification

Problem ID: SIH25001 Target Users: DHOs & Health Administrators Framework: React (TypeScript) + Vite

1. Frontend Technology Stack & Library Selection

LAYER	SELECTED LIBRARY	ARCHITECTURAL PURPOSE
Framework	React.js (TypeScript + Vite)	Provides strict type safety for complex epidemiological JSON responses and rapid build bundling.
UI Styling	Tailwind CSS + Shadcn UI	Enterprise administrative layout with dark/light theme toggles and accessible components.
GIS Mapping	Leaflet.js (react-leaflet + leaflet.heat)	Renders spatial vector heatmaps and village boundaries using OpenStreetMap tiles with zero API cost.
Data Visualization	Recharts / Tremor	Renders responsive SVG charts for time-series disease trend forecasting and correlation analytics.
State & Query Sync	TanStack Query (React Query)	Handles asynchronous data fetching, background polling, and cache invalidation from FastAPI.
Real-Time Alerts	WebSocket API / Socket.io-client	Pushes live notification banners when ASHA workers log critical red-flag symptoms in high-risk zones.
Export Engine	jsPDF + SheetJS (xlsx)	Generates client-side downloadable PDF epidemiological bulletins and CSV telemetry dumps.

2. Detailed Module Feature Specifications

MODULE 1 Interactive Spatio-Temporal GIS Map (Main Command View)

- **7–14 Day Predictive Outbreak Heatmap:** Renders color-coded risk vectors dynamically:
 - **GREEN (RISK < 0.35)** Low baseline vector activity.
 - **AMBER (0.35 ≤ RISK < 0.70)** Moderate warning alert zone.
 - **RED (RISK ≥ 0.70)** High outbreak probability predicted by Model A (LSTM).
- **Health Infrastructure Overlay:** Interactive markers for PHCs and CHCs displaying available bed capacity, ORS stock counts, and on-duty doctors.
- **Temporal Time-Scrubber:** Playback slider allowing administrators to inspect 30-day historical logs and scrub forward 14 days into predictive forecast horizons.

MODULE 2 Outbreak Predictive Analytics Hub

- **Multi-Axis Time-Series Plotter:** Graphs actual cases vs. predicted incidence alongside precipitation (mm) and relative humidity (%) to track vector incubator conditions.
- **Automated Anomaly Detection:** Highlights statistical spikes when daily symptom logs cross threshold levels via Z-score evaluation ($Z > 2.5$).
- **Demographic Breakdown:** Categorizes patient risk by age brackets (<5 yrs, 5–18, 18–60, 60+) and symptom clusters (Fever, Dehydration, Vomiting).

MODULE 3 ASHA Field Telemetry Stream

- **Real-Time Log Stream:** Sortable, paginated table of incoming mobile logs with status tags (Green / Amber / Red).
- **Sync Status Audit:** Tracks whether records arrived via instant online upload or batched SQLite offline sync.
- **Multi-Param Filters:** Filter records by District, PHC region, date range, or specific symptom flags.

MODULE 4 Resource Allocation & Dispatch

- **PHC Supply Monitor:** Live tracking of essential medical inventory (ORS, IV Ringer's Lactate, Paracetamol) with low-stock alerts (<50 units).
- **SMS / WhatsApp Broadcast:** One-click trigger to send emergency operational warnings to local medical staff when a village reaches Red status.

MODULE 5 RAG & Guidelines Manager

- **Document Vector Ingestion:** Drag-and-drop interface for uploading official WHO/IDSP medical PDFs directly into Qdrant Vector DB.
- **RAG Playground:** Testing sandbox for admins to query the LLM context pipeline prior to field deployment.

MODULE 6 Governance & Compliance Reports

- **One-Click PDF Bulletin:** Formats district epidemiological reports featuring current map state, top risk clusters, and resource metrics.
- **Raw Data Exporters:** Secure options to download anonymized datasets in CSV, Excel, and JSON formats.

3. Application Navigation & Component Architecture

Client-Side Viewport Routing Diagram

```
/ (Dashboard Layout Container)
├─ Navbar Header (District Selector | Alert Notification Bell | System Health Status)
├─ Sidebar Navigation
│   ├── 🗺️ GIS Spatial Map (/gis-map) [Default Landing]
│   ├── 📊 Outbreak Analytics & Forecast (/analytics)
│   ├── 📄 ASHA Symptom Telemetry (/field-telemetry)
│   ├── 📦 Inventory & Emergency Dispatch (/resource-management)
│   ├── 📖 RAG Medical Base Admin (/rag-admin)
│   └── 📄 Automated Reports Generator (/reports)
└─ Main Content Container (Dynamic Route Viewport)
```

React Component File Tree Hierarchy

```
src/
├─ components/
│   ├── common/           # Navbar, Sidebar, StatusBadge, Modal, DataTable
│   ├── map/
│   │   ├── MapContainer.tsx # React-Leaflet base map initialization
│   │   ├── RiskHeatmap.tsx  # Dynamic Leaflet.heat overlay layer
│   │   └── PHCMarkers.tsx   # Custom GeoJSON facility pins
│   ├── charts/
│   │   ├── TrendChart.tsx  # Recharts multi-axis line & bar charts
│   │   └── AnomalyCard.tsx  # Statistical Z-score alert components
│   └── telemetry/
│       └── LogTable.tsx     # Virtualized data grid for incoming logs
├─ services/
│   ├── api.ts             # Axios client instance connecting to FastAPI
│   └── websocket.ts       # Socket.io handler for push alerts
└─ store/
    └── useFilterStore.ts   # Zustand global store for state filtering
```